

Session 4

Technical Drawings, Exploded View & BOM

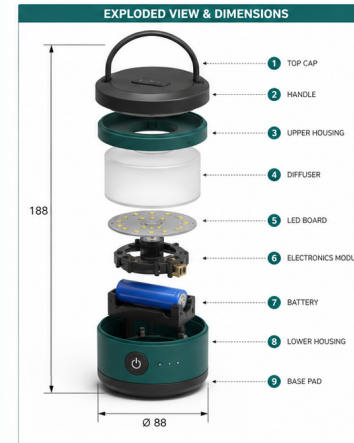
Brief and practical checklist for what each team must create.

Convert the product into parts and assemblies.

Define what is manufactured vs. purchased.

Prepare the first BOM and drawing checklist.

TECHNICAL DRAWINGS, EXPLODED VIEW AND BOM



BILL OF MATERIALS (BOM)					
ITEM	PART NAME	MATERIAL	QTY	UNIT COST	TOTAL COST
1	Top Cap	ABS	1	2.10	2.10
2	Handle	TPE	1	1.20	1.20
3	Upper Housing	ABS	1	1.80	1.80
4	Diffuser	PC	1	1.50	1.50
5	LED Board	Aluminum	1	1.70	1.70
6	Electronics Module	—	1	3.60	3.60
7	Battery (18650)	Li-ion	1	2.90	2.90
8	Lower Housing	ABS	1	2.20	2.20
9	Base Pad	Silicone	1	0.60	0.60
10	Screws (M2.5)	Steel	6	0.05	0.30
11	Gasket	Silicone	1	0.30	0.30
12	Charging Port (USB-C)	—	1	0.70	0.70
13	On/Off Button	Silicone	1	0.20	0.20
TOTAL					17.10

12 UNIQUE PARTS
6 ASSEMBLY FASTENERS

EST. MATERIAL COST
17.10

EST. WEIGHT
312 g



Today: translate design intent into parts, drawings and a preliminary BOM.

Output: a report fragment ready to become part of the final technical report.

Session goal

What must solve today.

What information would a manufacturer need to understand and produce your product?

1

1. Identify parts

List every piece, module, connector, textile piece, hardware item and consumable.

2

2. Organize assembly

Show how parts connect, open, fold, attach or are replaced.

3

3. Build the BOM

Create a preliminary list with quantities, materials, processes, suppliers and costs.

What to submit for Session 4

Keep it short, clear and advanced enough to integrate into the final report.

1 Product architecture

Parts, modules and how they work together.

3 Exploded view

Sketch, CAD screenshot or diagram with labels.

5 Preliminary BOM

Quantity, material, process, supplier and estimated cost.

7 Risks + missing info

What still needs measurement, testing or supplier confirmation.

2 Parts breakdown

Manufactured, commercial and consumable parts.

4 Drawing checklist

Which drawings are needed for the final technical report.

6 Assembly sequence

Basic order of construction or integration.

Parts breakdown

Make the product visible as a system of pieces.

Manufactured parts

Custom pieces made for your project:

- shells / bodies
- textile or leather patterns
- molded / printed components
- custom pads or covers

Commercial parts

Purchased components:

- buckles / zippers
- screws / magnets
- straps / webbing
- electronics / batteries
- hooks / latches

Consumables + finishes

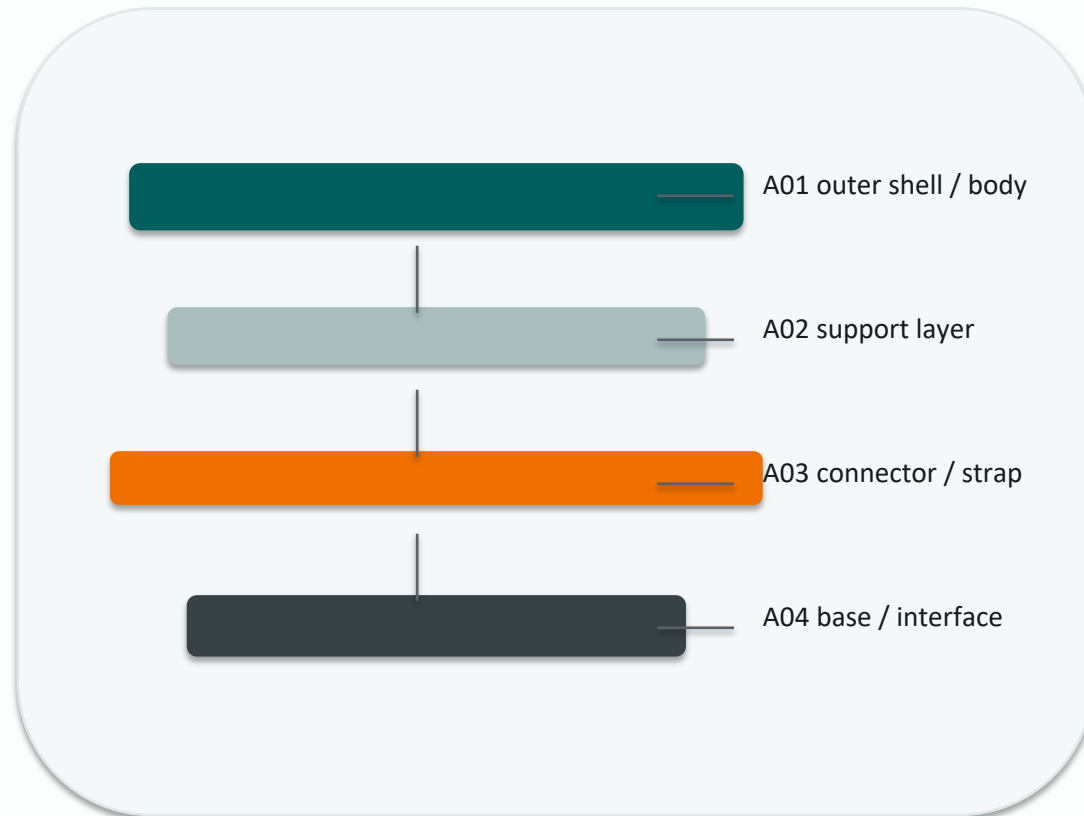
Materials used during production:

- adhesives
- thread
- coatings
- labels
- paint / dye
- packaging

If a part exists in the product, it must appear in the parts list and later in the BOM.

Exploded view or assembly diagram

A sketch is acceptable if it clearly explains the product.



1

Must show

All main parts separated and labeled.

2

Must connect

Each label must match the BOM row.

3

Must clarify

How parts attach, move, fold, open or are replaced.

Preliminary BOM

The BOM is the inventory of your product.

Part / Component	Qty.	Type	Material	Process / Supplier	Cost	Function	Status
Main shell / body	1	Manufactured	To define	3D print / injection / sewing	TBD	Main structure	needs dimensions
Strap / support	2	Commercial / manuf.	Nylon / textile	Supplier + stitching	TBD	Adjustment	needs test
Buckle / latch	1	Commercial	POM / metal	Supplier	TBD	Closure	choose type
Fasteners / thread	Several	Consumable	Steel / thread	Supplier	TBD	Assembly	count needed

Use “TBD / needs evidence” only when necessary. Do not leave empty cells.

Minimum: 10–15 rows depending on product complexity.

Drawing checklist

Do not need perfect final drawings, just must know what to create.

General product drawing	Complete product + main dimensions
Part drawings	Individual pieces with dimensions
Assembly drawing	How parts fit together
Exploded view	Separated parts + labels
Detail drawing	Closures, joints or critical zones
Pattern / flat layout	Textile, leather, fabric or sheet products
Electronic diagram	Only if sensors, batteries or wiring exist

What most teams need to improve

Based on the last submissions: be concrete and measurable.

1 Dimensions

Do not write only “to be defined”. Identify which dimensions are missing and why they matter.

2 Suppliers + costs

Start searching possible suppliers, commercial parts and price ranges.

3 Evidence

Plan tests: load, abrasion, water, magnets, comfort, assembly time or movement.

4 Clear diagrams

Include at least one labeled exploded view, pattern, sketch or assembly map.

5 Parts table

Every material decision must connect to a specific part or component.

6 Separate language

Technical report = production. Visual report = user understanding and purchase.

NotebookLM prompt for Session 4

Students can copy this and adapt it to their project.

Using only the uploaded sources, create a Session 4 report titled "Technical Drawings, Exploded View and Preliminary BOM".

Include: product architecture, parts breakdown, manufactured parts, commercial parts, exploded view description, drawing checklist, preliminary BOM, assembly sequence, technical risks, missing information and next steps.

Create a BOM table with: Part / Component, Quantity, Type, Material, Process or Supplier, Estimated Cost, Function and Status.

Do not invent final data. If information is missing, write "to be defined", "needs measurement" or "needs evidence".

End-of-class output

By the end of Session 4, each team should leave with these four things.

1. Parts list

All pieces, modules, commercial parts and consumables.

2. First exploded view

Labeled sketch or CAD screenshot showing how parts connect.

3. Preliminary BOM

Rows with quantity, material, process, supplier/cost and status.

4. Drawing plan

List of drawings needed before the final technical report.

Session 4 is not about making perfect plans yet. It is about knowing exactly what must be drawn, bought, made and tested.